

№7. - : M/M/c

2026-09-02

1

2

3

4

5

 $M/M/c$

6

7

1



-
-

-01-23



-
-
-

-01-23
: 1032225099



-
-
-
-

-01-23
: 1032225099



- 01-23
: 1032225099



-
- -01-23
- : 1032225099
-
-
- : M/M/c
- : Julia, DrWatson, CSV, DataFrames, CairoMakie,
Literate.jl



2



-

M/M/c

- - M/M/c
- M/M/c

- - M/M/c
- M/M/c
-

- - M/M/c
- M/M/c
-
-

- - $M/M/c$
- $M/M/c$
-
-
- ,

1

Julia-

DrWatson

- 1 Julia- DrWatson
- 2 src/QueueingModels.jl

- 1 Julia- DrWatson
- 2 src/QueueingModels.jl
- 3

- 1 Julia- DrWatson
- 2 src/QueueingModels.jl
- 3
- 4 CSV-

- ① Julia- DrWatson
- ② src/QueueingModels.jl
- ③
- ④ CSV-
- ⑤ clean, markdown ipynb-

3

:

:

- λ —

$$\rho = \frac{\lambda}{c\mu}$$

$$\rho \geq 1$$

.

:

- λ —
- μ —

:

$$\rho = \frac{\lambda}{c\mu}$$

$$\rho \geq 1$$

.

:

- λ —
- μ —
- c —

$$\rho = \frac{\lambda}{c\mu}$$

:

$$\rho \geq 1$$

.

:

- λ —
- μ —
- c —

$$\rho = \frac{\lambda}{c\mu}$$

:

•

P_{wait}

$\rho \geq 1$

.

:

- λ —
- μ —
- c —

$$\rho = \frac{\lambda}{c\mu}$$

:

- P_{wait}
- W_q

$\rho \geq 1$.

:

- λ —
- μ —
- c —

$$\rho = \frac{\lambda}{c\mu}$$

:

- P_{wait}
- W_q
- L_q

$\rho \geq 1$.

:

- λ —
- μ —
- c —

$$\rho = \frac{\lambda}{c\mu}$$

:

- P_{wait}
- W_q
- L_q
- L
- $\rho \geq 1$

• N

T_k.

:

:

- N
- S
- repairers

T_k.

: :

- N
- S
- repairers
- tau_f

T_k.

:

:

- N
- S
- repairers
- tau_f
- tau_r

T_k.

:

- N
- S
- repairers
- tau_f
- tau_r

:

•

T_k.

:

- N
- S
- repairers
- tau_f
- tau_r

:

-
-

T_k.

:

- N
- S
- repairers
- tau_f
- tau_r

:

-
-
-

T_k.

:

- N
- S
- repairers
- tau_f
- tau_r

:

-
-
-
-

T_k.

4

:

- src/QueueingModels.jl

:

```
cd ~/labs/lab07/project
~/juliaup/bin/julia --project=. -e 'us
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. script
```

:

- src/QueueingModels.jl
- scripts/mmc.jl

:

```
cd ~/labs/lab07/project
~/juliaup/bin/julia --project=. -e 'us
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. script
```


:

- src/QueueingModels.jl
- scripts/mmc.jl
- scripts/mmc_parameters.jl

:

```
cd ~/labs/lab07/project
~/juliaup/bin/julia --project=. -e 'us
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. script
```

:

- src/QueueingModels.jl
- scripts/mmc.jl
- scripts/mmc_parameters.jl
- scripts/ross.jl

:

```
cd ~/labs/lab07/project
~/juliaup/bin/julia --project=. -e 'us
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. script
```

:

- src/QueueingModels.jl
- scripts/mmc.jl
- scripts/mmc_parameters.jl
- scripts/ross.jl
- scripts/ross_parameters.jl

:

```
cd ~/labs/lab07/project
~/juliaup/bin/julia --project=. -e 'us
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. -e 'in
~/juliaup/bin/julia --project=. script
```

M/M/c

```
mabaptishta@fedora: /work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07$ cd project
mabaptishta@fedora: /work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ./
.juliaup/bin/julia --project=, -e 'using Pkg; Pkg.instantiate()'
```

Installing known registries into ~/.julia

Added 'General' registry to ~/.julia/registries

Project No packages added to or removed from '~/.work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project/,Project.toml'

Manifest No packages added to or removed from '~/.work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project/,Manifest.toml'

```
modeling/labs/lab07/project/,Manifest.toml
mabaptishta@fedora: /work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ./
.juliaup/bin/julia --project=, -e 'include("scripts/mmc.jl")'
```

M/M/c.

customers =

5000, mean wait = 8.3449, utilization = 0.8989.

```

modeling/labs/lab07/project/; ssh root@torc
mabaptishta@fedora:~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ -/
.juliaup/bin/julia --project=, -e 'include("scripts/mmc.jl")'
ERROR: ArgumentError: opening file "/home/mabaptishta/work/2026-1--study-simulation--modeling/2026-1--study-simulation

```



18

```
modeling/labs/lab07/project/; ssh root@torc
mabaptishta@fedora:~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ~/
.juliaup/bin/julia --project=, -e 'include("scripts/mmc.jl")'
ERROR: ArgumentError: opening file "/home/mabaptishta/work/2026-1--study-simulation--modeling/2026-1--study-simulation--
```

- 18
- : lambda in [0.3, 0.6, 0.9]

```
modeling/labs/lab07/project/; ssh root@torc
mabaptishta@fedora:~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ~/
.juliaup/bin/julia --project=, -e 'include("scripts/mmc.jl")'
ERROR: ArgumentError: opening file "/home/mabaptishta/work/2026-1--study-simulation--modeling/2026-1--study-simulation--
```

- 18
- : lambda in [0.3, 0.6, 0.9]
- : c in 1:6

```

modeling/labs/lab07/project/; ssh root@torc
mabaptishta@fedora:~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ~/
.juliaup/bin/julia --project=, -e 'include("scripts/mmc.jl")'
ERROR: ArgumentError: opening file "/home/mabaptishta/work/2026-1--study-simulation--modeling/2026-1--study-simulation

```

- 18
- : lambda in [0.3, 0.6, 0.9]
- : c in 1:6
-

mmc_parameter_scan.csv

M/M/c:

:

•

;

M/M/c:

:

•

;

•

;

M/M/c:

:

- ;
- ;
- ;

M/M/c:

:

- ;
- ;
- ;
- ;

M/M/c:

:

- ;
- ;
- ;
- ;
- time-weighted .

M/M/c:

- ;
- ;
- ;
- ;
- time-weighted .

:

;

M/M/c:

- ;
- ;
- ;
- ;
- time-weighted .

:

- ;
- ;

M/M/c:

- ;
- ;
- ;
- ;
- time-weighted .

:

- ;
- ;
- ;

M/M/c:

- ;
- ;
- ;
- ;
- time-weighted .

:

- ;
- ;
- ;
- ;

M/M/c:

- ;
- ;
- ;
- ;
- time-weighted .

:

- ;
- ;
- ;
- ;
- .

5

M/M/c

:

- $\lambda = 0.9$

:

:

- $\lambda = 0.9$
- $\mu = 0.5$

:

:

- $\lambda = 0.9$
- $\mu = 0.5$
- $c = 2$

:

:

- `lambda = 0.9`
- `mu = 0.5`
- `c = 2`
- `num_customers = 5000`

:

```
:  
• lambda = 0.9  
• mu = 0.5  
• c = 2  
• num_customers = 5000  
• seed = 123  
:
```


- :
- $\lambda = 0.9$
- $\mu = 0.5$
- $c = 2$
- $\text{num_customers} = 5000$
- $\text{seed} = 123$
- :
- $\text{sim_wq} = 8.3449$

:

- $\lambda = 0.9$
- $\mu = 0.5$
- $c = 2$
- $\text{num_customers} = 5000$
- $\text{seed} = 123$

:

- $\text{sim_wq} = 8.3449$
- $\text{analytic_wq} = 8.5263$

:

- $\lambda = 0.9$
- $\mu = 0.5$
- $c = 2$
- $\text{num_customers} = 5000$
- $\text{seed} = 123$

:

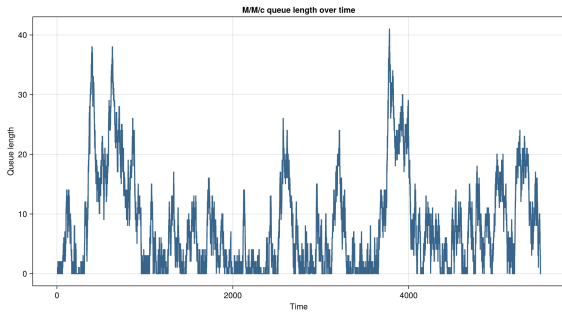
- $\text{sim_wq} = 8.3449$
- $\text{analytic_wq} = 8.5263$
- $\text{sim_pwait} = 0.8556$

:

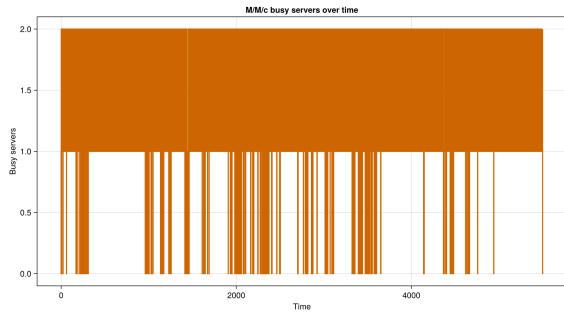
- $\lambda = 0.9$
- $\mu = 0.5$
- $c = 2$
- $\text{num_customers} = 5000$
- $\text{seed} = 123$

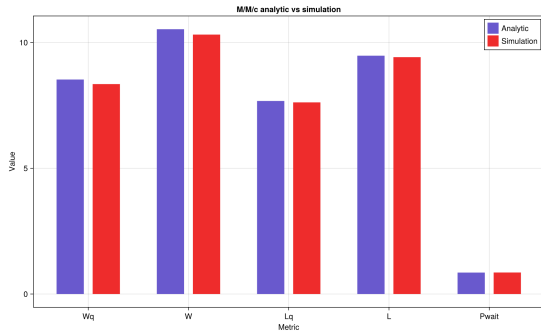
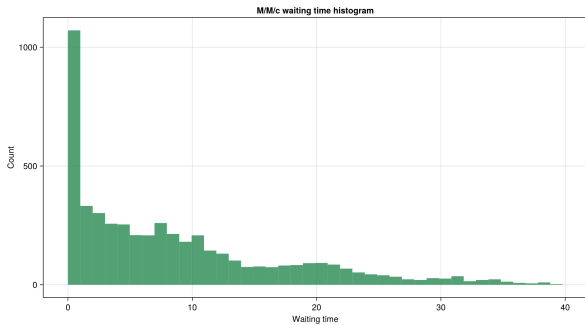
:

- $\text{sim_wq} = 8.3449$
- $\text{analytic_wq} = 8.5263$
- $\text{sim_pwait} = 0.8556$
- $\text{sim_utilization} = 0.8989$



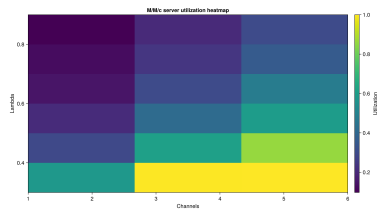
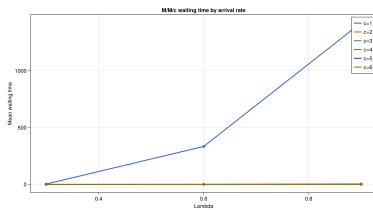
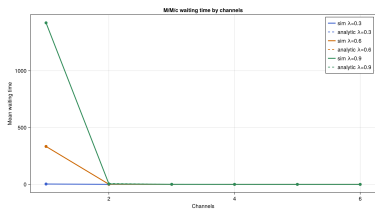
$\rho = 0.9$





W_q L_q ,

M/M/c



lambda

W_q ,

lambda

$c = 1$

6

:

- $N = 10$

:

:

- $N = 10$
- $S = 3$

:

:

- $N = 10$
- $S = 3$
- $\text{repairers} = 1$

:

:

- `N = 10`
- `S = 3`
- `repairers = 1`
- `runs = 300`

:

:

- $N = 10$
- $S = 3$
- $\text{repairers} = 1$
- $\text{runs} = 300$
- $\text{tau}_f = 100.0$

:

```
:  
• N = 10  
• S = 3  
• repairers = 1  
• runs = 300  
• tau_f = 100.0  
• tau_r = 1.0  
:
```

```
:  
• N = 10  
• S = 3  
• repairers = 1  
• runs = 300  
• tau_f = 100.0  
• tau_r = 1.0  
:  
• mean_crash_time = 11937.8165
```

```
:  
• N = 10  
• S = 3  
• repairers = 1  
• runs = 300  
• tau_f = 100.0  
• tau_r = 1.0  
:  
• mean_crash_time = 11937.8165  
• analytic_crash_time = 12340.0
```



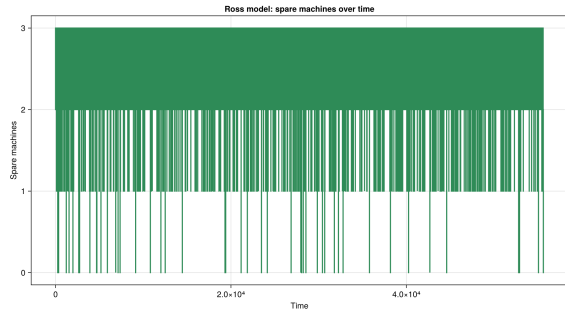
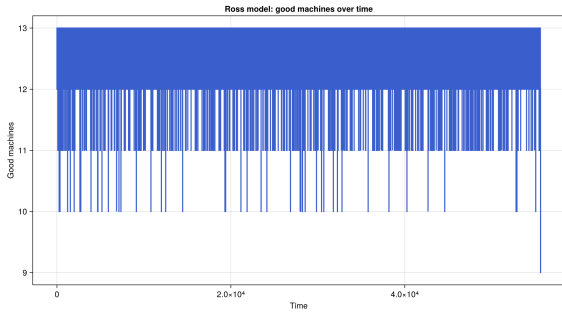
```
:  
• N = 10  
• S = 3  
• repairers = 1  
• runs = 300  
• tau_f = 100.0  
• tau_r = 1.0  
:  
• mean_crash_time = 11937.8165  
• analytic_crash_time = 12340.0  
• mean_repair_queue = 0.0120
```

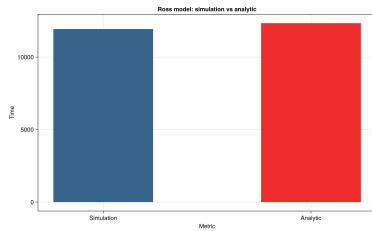
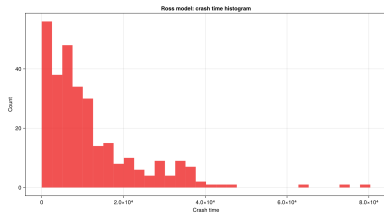
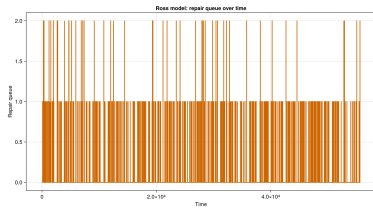
```
:  
• N = 10  
• S = 3  
• repairers = 1  
• runs = 300  
• tau_f = 100.0  
• tau_r = 1.0  
:  
• mean_crash_time = 11937.8165  
• analytic_crash_time = 12340.0  
• mean_repair_queue = 0.0120  
• repairer_utilization = 0.1010
```

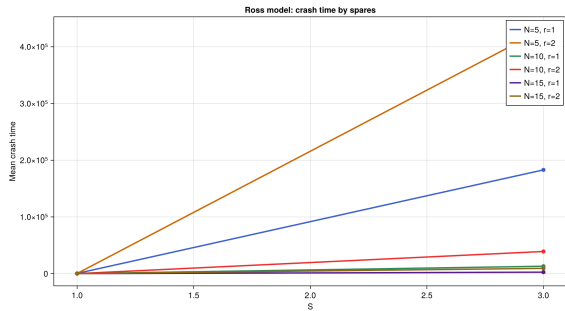
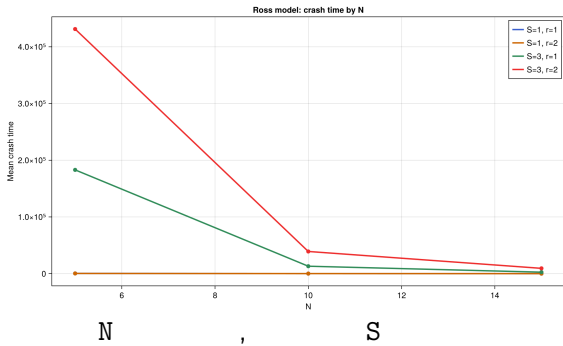
```
habaptishta@fedora: ~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ~/
.juliaup/bin/julia --project=, -e 'include("scripts/ross.jl")'
Activating project at ~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/proje
```

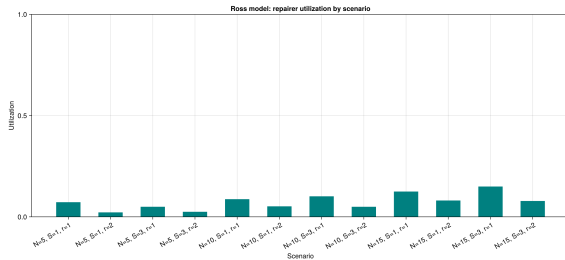
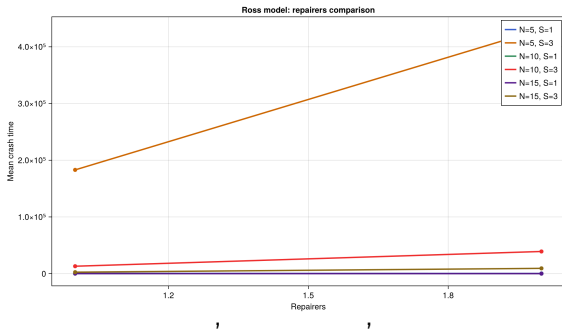
```
/labs/lab07/project/scripts/ross.jl:4
habaptishta@fedora: ~/work/2026-1--study-simulation--modeling/2026-1--study-simulation--modeling/labs/lab07/project$ ~/
.juliaup/bin/julia --project=, -e 'include("scripts/ross__parameters.jl")'
```

12









7

scripts/tangle.jl :

- mmc_clean.jl, mmc.md, mmc.ipynb

scripts/tangle.jl :

- mmc_clean.jl, mmc.md, mmc.ipynb
- mmc_parameters_clean.jl, mmc_parameters.md, mmc_parameters.ipynb

scripts/tangle.jl :

- mmc_clean.jl, mmc.md, mmc.ipynb
- mmc_parameters_clean.jl, mmc_parameters.md, mmc_parameters.ipynb
- ross_clean.jl, ross.md, ross.ipynb

scripts/tangle.jl :

- mmc_clean.jl, mmc.md, mmc.ipynb
- mmc_parameters_clean.jl, mmc_parameters.md, mmc_parameters.ipynb
- ross_clean.jl, ross.md, ross.ipynb
- ross_parameters_clean.jl, ross_parameters.md, ross_parameters.ipynb

```
[ Info: generating plain script file from '~/labs/lab07/project/scripts/mmc_literate.jl'
[ Info: writing result to '~/labs/lab07/project/scripts/mmc_clean.jl'
[ Info: generating notebook from '~/labs/lab07/project/scripts/mmc_literate.jl'
[ Info: executing notebook 'mmc.ipynb'
[ Info: writing result to '~/labs/lab07/project/notebooks/mmc.ipynb'
[ Info: generating markdown page from '~/labs/lab07/project/scripts/mmc_literate.jl'
[ Info: writing result to '~/labs/lab07/project/docs/mmc.md'
generated for mmc
[ Info: generating plain script file from '~/labs/lab07/project/scripts/mmc_parameters_literate.jl'
[ Info: writing result to '~/labs/lab07/project/scripts/mmc_parameters_clean.jl'
[ Info: generating notebook from '~/labs/lab07/project/scripts/mmc_parameters_literate.jl'
[ Info: executing notebook 'mmc_parameters.ipynb'
[ Info: writing result to '~/labs/lab07/project/notebooks/mmc_parameters.ipynb'
[ Info: generating markdown page from '~/labs/lab07/project/scripts/mmc_parameters_literate.jl'
[ Info: writing result to '~/labs/lab07/project/docs/mmc_parameters.md'
generated for mmc_parameters
[ Info: generating plain script file from '~/labs/lab07/project/scripts/ross_literate.jl'
[ Info: writing result to '~/labs/lab07/project/scripts/ross_clean.jl'
[ Info: generating notebook from '~/labs/lab07/project/scripts/ross_literate.jl'
[ Info: executing notebook 'ross.ipynb'
[ Info: writing result to '~/labs/lab07/project/notebooks/ross.ipynb'
[ Info: generating markdown page from '~/labs/lab07/project/scripts/ross_literate.jl'
[ Info: writing result to '~/labs/lab07/project/docs/ross.md'
generated for ross
[ Info: generating plain script file from '~/labs/lab07/project/scripts/ross_parameters_literate.jl'
[ Info: writing result to '~/labs/lab07/project/scripts/ross_parameters_clean.jl'
[ Info: generating notebook from '~/labs/lab07/project/scripts/ross_parameters_literate.jl'
[ Info: executing notebook 'ross_parameters.ipynb'
```

```
~/labs/lab07/project$ ls
..  data
mmc_customers.csv
mmc_events.csv
mmc_parameter_scan.csv
mmc_summary.csv
ross_events_sample.csv
ross_parameter_scan.csv
ross_runs.csv
ross_summary.csv
```

tangle.jl clean- , md- ipynb- .
data, plots, docs notebooks.



M/M/c

-
-

M/M/c

λ c

-
-
-

M/M/c

λ c

-
-
-
-

M/M/c

λ c

-
-
-
-
-

M/M/c

λ c

, , ,